Maths 3 _ 5 Mai 2026.

Exo 1 : (E_1) ED linéaire d'ordre 2 à coef constant.

$P(X) = 4X^2 - 4X + 1 = (2X-1)^2$

Sol de l'EDLSSM (E_1') : $(K_1 + x K_2) e^{\frac{1}{2}x}$

On remarque que 4 est solution

Solution de l'équation différentielle avec second membre

$y(x) = (K_1 + x K_2) e^{x/2} + 4$ $\quad y'(x) = \left(\frac{K_1}{2} + K_2 + \frac{1}{2} x K_2\right) e^{x/2}$

$y(0) = 0 \iff K_1 = -4$ $\quad y'(0) = \frac{K_1}{2} + K_2$

Donc sol. de (E_1) $y(x) = 4 + (2x-4) e^{x/2}$

(E_2) EDL linéaire d'ordre 1 à coef constant.

$\begin{cases} u = x + ay \\ v = y \end{cases}$ $\quad \varphi(x,y) = \Gamma(u,v) = \Gamma(x+ay ; y)$

$\frac{\partial \varphi}{\partial x} = \frac{\partial \Gamma}{\partial u}$ $\quad \frac{\partial \varphi}{\partial y} = a\frac{\partial \Gamma}{\partial u} + \frac{\partial \Gamma}{\partial v}$

"E_2" $\iff 3\frac{\partial \Gamma}{\partial u} + a\frac{\partial \Gamma}{\partial u} + \frac{\partial \Gamma}{\partial v} = \Gamma$ $\quad$ Posons $a = -3$

$\frac{\partial \Gamma}{\partial v} = \Gamma$ donc $\Gamma(u,v) = K(u) e^v$ avec $K : \mathbb{R} \to \mathbb{R}$ quelconque

D'où les solutions sont $\varphi(x,y) = K(x-3y) e^y$ avec $K : \mathbb{R} \to \mathbb{R}$ quelconque

$\varphi(e^t, 0) = t \iff K(e^t) e^0 = t \iff K(z) = \ln z$. $\forall z \in \mathbb{R}_+^*$

$\varphi(x,y) = \ln(x-3y)\, e^y$: Solution de (E_2).

Exo 2 : $\Gamma = \text{grad}(\text{div}\,\Phi) - \text{rot}(\text{rot}\,\Phi)$

$$\Gamma_x = \frac{\partial}{\partial x}\left(\frac{\partial \Phi_1}{\partial x} + \frac{\partial \Phi_2}{\partial y} + \frac{\partial \Phi_3}{\partial z}\right) - \left(\begin{pmatrix} \frac{\partial}{\partial y}\frac{\partial \Phi_2}{\partial x} - \frac{\partial}{\partial z}\frac{\partial \Phi_1}{\partial x} \\ \frac{\partial}{\partial z}\frac{\partial \Phi_1}{\partial y} - \frac{\partial}{\partial x}\frac{\partial \Phi_2}{\partial y} \\ \frac{\partial}{\partial x}\frac{\partial \Phi_3}{\partial z} - \frac{\partial}{\partial y}\frac{\partial \Phi_1}{\partial z} \end{pmatrix} \wedge \begin{pmatrix} \frac{\partial \Phi_3}{\partial y} - \frac{\partial \Phi_2}{\partial z} \\ \frac{\partial \Phi_1}{\partial z} - \frac{\partial \Phi_3}{\partial x} \\ \frac{\partial \Phi_2}{\partial x} - \frac{\partial \Phi_1}{\partial y} \end{pmatrix}\right)_x$$

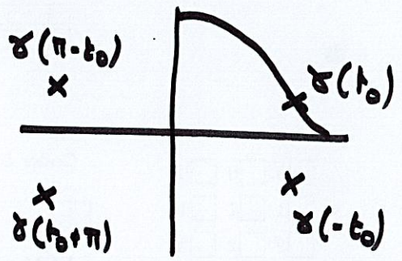
$$= \frac{\partial^2 \Phi_1}{\partial x^2} + \frac{\partial^2 \Phi_2}{\partial x \partial y} + \frac{\partial^2 \Phi_3}{\partial x \partial z} - \frac{\partial^2 \Phi_2}{\partial y \partial x} + \frac{\partial^2 \Phi_1}{\partial y^2} + \frac{\partial^2 \Phi_1}{\partial z^2} - \frac{\partial^2 \Phi_3}{\partial z \partial x}$$

$$= \frac{\partial^2 \Phi_1}{\partial x^2} + \frac{\partial^2 \Phi_1}{\partial y^2} + \frac{\partial^2 \Phi_1}{\partial z^2} = \Delta \Phi_1$$

Exo 3 : Lorsque l'on passe de $\gamma(t)$ à $\gamma(-t)$ on effectue une symétrie /(Ox)

$t \to t+\pi$ Symétrie centrale de centre O

$t \to \pi - t$ ——— d'axe /(Oy).



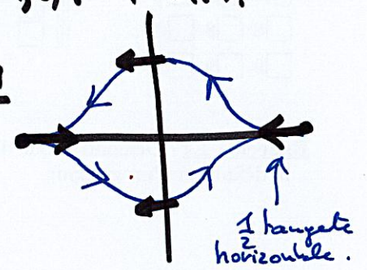
3.2 : $\gamma'(t) = (-\sin t ; 3\cos t \sin^2 t)$ $\quad \gamma'(\frac{\pi}{2}) = (-1, 0)$ vecteur horizontale

3.3 : $\gamma'(0) = 0$ $\quad \gamma''(t) = (-\cos t ; 6\cos^2 t \sin t - 3\sin^3 t)$ $\quad \gamma''(0) = (-1, 0)$.

$\gamma(t) = \gamma(0) + \gamma'(0)t + \frac{1}{2}\gamma''(0)t^2 + o(t^2) = (1,0) + \frac{1}{2}(-1,0)t^2 + o(t^2)$.

Demi tangente horizontale

3.4



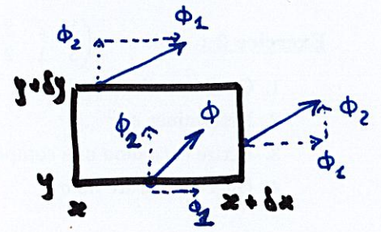
3.5 : $\text{Aire}(K) = \iint_K 1 \, dx\,dy$.

prenons $Q = x$ et $P = 0$.

G.R : $\iint_K \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\, dx\,dy = \int_{\partial K^+} P\,dx + Q\,dy$.

$$A = \int_\gamma x\,dy = \int_0^{2\pi} \cos t \left(3\cos t \sin^2 t\right) dt = 3\int_0^{2\pi} \cos^2 t \sin^2 t\, dt = \frac{3}{4}\int_0^{2\pi} (2\cos t \sin t)^2\, dt$$

$$= \frac{3}{4}\int_0^{2\pi} (\sin(2t))^2\, dt = \frac{3\pi}{4}.$$



Exo 4 : Ce qui sort du côté du bas

$-\Phi_2(\tilde{x}; y)\,\delta x$

du côté de droite : $\Phi_1(x+\delta x; \tilde{y})\,\delta y$

du côté du haut : $\Phi_2(\tilde{\tilde{x}}; y+\delta y)\,\delta x$

——— gauche : $-\Phi_1(x; \tilde{\tilde{y}})\,\delta y$.

La somme des 4 côtés,

$$\underbrace{\left(\Phi_1(x+\delta x; \tilde{y}) - \Phi_1(x; \tilde{y})\right)}_{\frac{\partial \Phi_1}{\partial x}(\ddot{x}; \tilde{y})\,\delta x}\delta y + \underbrace{\left(\Phi_2(\tilde{\tilde{x}}, y+\delta y) - \Phi_2(\tilde{\tilde{x}}, y)\right)}_{\frac{\partial \Phi_2}{\partial y}(\tilde{\tilde{x}}; \ddot{y})\,\delta y}\delta x$$

Lorsque $\delta x \to 0$ et $\delta y \to 0$ $\quad \ddot{x} \to x$ $\quad \ddot{y} \to y$; $\tilde{\tilde{x}} \to x$ $\quad \tilde{\tilde{y}} \to y$

D'où on obtient $\simeq \left(\frac{\partial \Phi_1}{\partial x}(x,y) + \frac{\partial \Phi_1}{\partial y}(y,y)\right) \underbrace{\delta x \delta y}_{\text{aire du rectangle}}$.